Mangroves provide a unique habitat mainly for many crustaceans and mollusks.

Mangroves are found in Puttalam, Batticaloa, Trincomalee and Galle.Bentota and Negambo.

Figure 8.20 : Mangroves

Salt marshes

These are marshlands restricted to the arid coastal regions of the country where soil dries up to form crystals of salts during the dry season. Low rainfall, high wind, high temperatures and loose sand blowing with salt are some of the major characteristics found in this ecosystem.

The vegetation has only few plant and animal species . Plants are short, contain fleshy succulent plant bodies One common plant species is *Salicornia* sp. Salt marshes are common in Puttalam, Mannar, Hambantota and Vakarai areas.

Figure 8.21 : Salt Marsh

Sea grass beds

In large lagoon areas with low wave action, the floor of the shallow sea is occupied by sea grasses. These are not grass species but appear like grasses due to the shape of leaves. E.g. Plant species such as *Halodule* spp and *Halophyla* spp. are common sea grasses found in Sri Lanka (especially from Kalpitiya to Mannnar. Sea grasses grow in a compact cluster and provides the sea bed a favourable habitat for breeding of many fish species. However, these areas are often disturbed by the fisheries activities as these are the parking areas of fishing boats.

Figure 8.22 : Sea grass bed

Coral reefs

Coral reefs are one of the natural wonders of the world. Coral reefs consist of calcareous structures secreted by a group of marine invertebrates. Coral reefs are famous for their spectacular beauty. They are considered as 'rain forests of the sea' because of their high productivity and high diversity of organisms inhabit them. Coral reefs can be seen in southern coast (Akurala to Tangalle), Gulf of Mannar, etc.

The reefs are habitats for a large number of fish species, invertebrates such as spiny lobsters, sea cucumbers, etc. Marine mammals and reptiles such as dolphins and sea turtles inhabit reefs occassionally.

Figure 8.23 : Coral reefs

Reservoirs

There are no natural lakes in Sri Lanka, but there are numerous ancient irrigation tanks mainly

scattered in the lowland dry zone. Typical irrigation tanks include 'Parakrama samudra', 'Kala wewa', 'Minneriya wewa and tissawewa.
Aquatic plant species commonly found in the reservoirs are

- S: Manel, E: Water lily (*Nymphaea* spp),
- S: Nelum. T: Tamarai (*Nelumbo nucifera*)
- S: Kekatiya, T: Koddi (*Aponogeton* spp).

Often free floating invasive alien plant species such as *Salvinia* and Japan jabara (S)/ water hyacinth (E) also can be seen in these tanks.

Figure8.24 : Reservoirs

Sea shore

The long sea shore of Sri Lanka varies in nature. The most common sea shore type is sandy sea shores. The sea shore areas share the characteristics of high temperature throughout, and salt spray and high winds especially during the monsoon seasons. Most of the sea shore plants have adapted to these conditions. Examples for these plants are Muhudu Binthamburu (S), Beach Morning Glory (E)- atampu (T) (*Ipomea pescaprae*), Maha rawana revula (S), Ravannan meesai (T)- *Spinifex littoreus* .

The vegetation gradually become stable a distance away from the tide mark, with the stabilization of the soil. In these areas plant species such as wara (S)/ erukkalai (T)- (*Calotropis gigantea*), Wetakeiya (S)/ talai (T)- *Pandanus* spp etc. can be found.

Figure 8.25 : Sea shore

Sand Dunes

Dunes are characterized by stunted or creping vegetation on large masses of sand. The sand dune structure is determined by wind speed and direction. Dunes are raised beaches of sand and are characteristic of certain coastal areas in the arid zone. Eg: near Mullativu, T

Figure 8.26 : Sand dunes

Biodiversity

Definitions

Biodiversity includes all forms of "life" on earth.

Biodiversity is the variability among living organisms from all sources including terrestrial, marine, and other aquatic ecosystems and their ecological interactions with the environment. Biodiversity is explained under three levels. They are genetic diversity, species diversity and ecosystem diversity.

a. Genetic diversity

The basic component of biological diversity is the genetic variation that exists both within and among species. This genetic variation is the basis for evolution.

b. Species diversity

This is simply the variation that can be recognized among different species. It includes the number of species (=species richness) and their abundance.

c. Ecosystem diversity is the variety of habitats, living communities and ecological processes in the living world.

Ecosystem diversity is the largest scale of biodiversity, and within each ecosystem, there is a great deal of both species and genetic diversity incorperated.

Ecosystem diversity on a global scale would be the variation in ecosystems in large regions (biomes) such as deserts, forests, grasslands, wetlands and oceans whereas in smaller localized regions it can be explained by means of different ecosystems.

The importance and values of biodiversity

The individual components of biodiversity—genes, species, and ecosystems provide the human society with a wide array of goods and services. Genes, species, and ecosystems of direct, indirect, or potential use to humanity are often referred to as "biological resources". Genes are used by plant breeders to develop new crop varieties. Many species are used as various foods, medicines, fibers, fuels and industrial products. This include food resources like grains, vegetables, fruits which are obtained from plant resources and meat, fish, egg, milk and milk products from animal resources. The biodiversity products can be harvested and consumed directly without passing through a formal market (non commercial goods). Example: fruits, fish, edible roots, leaves, nuts, flowers, meat, animal product like milk and honey, timber, firewood, fiber, wool, wax, resin, rubber, silk and decorative items, and traditional medicines, etc. Some products can be harvested and available through a formal market (commercial goods). Many industries such as food, textile, leather, silk, paper pulp are based on the direct use of biological resources. The ecosystems provide many services to us, such as air and water purification, erosion prevention and flood control.

Therefore the value of biodiversity is explained through its "goods" and "services" provided to humanity and sustenance of the environment.

- **Environmental service value**: This is the most important services provided by biodiversity in maintaining critical environmental functions. E.g. Carbon dioxide fixation through photosynthesis, maintaining of essential nutrition cycles, maintaining water cycle and recharging of ground water, soil formation and protection from erosion, regulating climate by recycling moisture into the atmosphere, water purification, pollination, etc.
- **Recreational value**: There is a great aesthetic value provided by biodiversity. Natural landscapes at undisturbed places are a delight to watch and also provide opportunities for recreational activities and hobbies such as bird watching, photography etc. Biodiversity provides inspiration in artistic activities like poetry, painting, dance etc. It promotes eco-tourism, helps to generate revenue by designing of zoological, botanical gardens, national parks etc.
- **Ethical value**: This is the right of all living beings to live on this planet, humans have no right to decide which species should exist since we are just a small part of the greater creation of nature.
- **Educational/Scientific value**: knowledge about biodiversity helps in new scientific discoveries and technological innovations to find solutions to the problems we face today. E.g. learning of other animals like nematodes, rats and primates has helped in understanding human body and development of medicines, knowledge about how animals react before a natural disaster is helpful in disaster management, Interacting with biodiversity is proven to be helpful in developing creativity, relieving stress and development of personality.
- **Social/Cultural/Religious values**: Biodiversity can be important to different societies and communities due to unique reasons E.g. Some wetland sites are sacred to Aborigines of Australia, twenty eight species of trees are sacred for Buddhists, Bulls are considered to be an important part of Hindu culture.

Figure 8.27 : Recreational value of BiodiversityLoss of biodiversity

Animal and plant species die off all the time. It's how the biological world rolls. However, things have changed dramatically in recent decades. According to some scientists, Earth is currently in the midst of its sixth mass extinction. The last major extinction occurred some 65 million years ago, when a large asteroid slammed off the coast of Mexico, killing off the dinosaurs and most everything else. Today, scientists say the extinction rate is as much as 1,000 times faster than what should be natural rate would be.This is solely because of the very high and negative impact of humans on biodiversity due to high population and development. Virtually all of Earth's ecosystems have been dramatically transformed through human actions, for example, many mangrove and coral reef areas have been lost. Within groups of conifers, cycads, amphibians, birds, and mammals up to 50% of species are threatened with extinction, according to the IUCN Red List. Within many species groups, such as amphibians, African mammals, and birds in agricultural lands, the majority of species have faced a decline in the size of their population, in their geographical spread, or both.

Threats to biodiversity

Habitat loss/fragmentation: Humans supplant natural ecosystems to grow food, harvest materials, and build our settlements. These actions alter or eliminate the conditions needed for plants and animals to survive. When natural habitats are converted into other human uses such as agriculture or built up area they are no longer able to support the species present in the original habitat. This result in the displacement or destruction of biodiversity. E.g. deforestation, filling of wetlands

Mass scale destruction of Mangrove in lagoons such as Negambo and Puttalam due to establishment of prawn culture destructed the biodiversity of mangroves in these areas.

Figure 8.28 : Habitat loss

When habitats are divided into fragments due to establishment of man built structures such as roads, the animals and plant species are forced to occupy smaller area in a crowded manner which is harder for biodiversity to sustain as in previous habitat conditions.

Overexploitation: Harvesting or exploiting biodiversity products in a manner and a rate which it cannot recover within the periods of exploitation leads to danger of biodiversity being completely lost.

E.g. Over collection of indigenous medicinal plants from forests in Sri Lanka for export such as Kotalahimbutu (S)/ - (*Salacia reticulata*). Export of sea cucumber for medicinal purposes from Sri Lankan shores. *Ebony*(E)/Kaluwara (S), Karun-kaali (T)- (*Diospyrus ebanum)* is threatened due to over exploitation during the colonial period. Ebony has a very slow growth rate and take many years to grow. Intense commercial fishing has led to over fishing threatening decline of food fish like Tuna and Cod in world's oceans.

Pollution: Pollution simply means addition of unwanted materials to air, water, soil. Due to extensive use of agrochemicals that wash away with rain water into the water bodies make the water rich in nutrients (eutrophication) resulting in algal blooms. Algal blooms create oxygen-depleted zone in aquatic ecosystems and greatly reduce the populations of fish and other aquatic species.

Uses of synthetic fertilizers for tea in montane areas also has resulted pollution of rivers in many down stream areas affecting the water quality and making it unsuitable for human use

Releasing of Sulphur dioxide (SO_2) and nitrous oxide (N_2O) gasses react with water and make the rain water acidic resulting acid rains. Acid rains caused by air pollution contributes to the death of trees killing many buds, leaves and the seedlings and causing damage to the plant roots.

Figure 8.29 : Water pollution affects aquatic biodiversity.

Introduction of invasive alien species: Invasive alien species are alien (exotic) plants and/or animals whose introduction and spread outside their natural geographic range threaten native biodiversity. Alien invasive species compete against or prey on native species, which can lead to their extinction. Once introduced , for a considerable period of time, they may lack natural predators in the new environment. This is a great opportunity for them to reproduces successfully and spread without limits to take over the environment. They can transport disease, out-compete native species, alter food chains, decrease biodiversity, and even change ecosystems properties by altering soil composition or creating habitats that encourage wildfires. E.g. Alien invasive species such as Lantana(E)/ Gandapana (S) / Nayunni (T)-(*Lantana camara*) does not allow germination and seedling growth of many other plants as it produces toxins which are added to soil through leaf litter. Extensive spread of Guinea grass(E)/ Gini- thana (S)/ Ginipullu (T) (*Panicum maximum*) in especially in the dry pathana areas facilitates fire due to its dry biomass during drought seasons.

Figure 8.30 : *Lantana* (Gandapana)

Climate change: Climate change is predicted to be the greatest long-term threat to biodiversity. Increasing temperatures and temperature extremes, increasingly severe droughts, rising sea levels, possible decrease in rainfall, regional flooding and reduced water availability change ecosystems. Many species will not be able to adapt themselves fast enough to keep up with the coming changes driving them to extinction or being endangered. Evidence suggests that the warming of the past century already has resulted in marked ecological changes, such as changes in growing seasons of crop species, distribution ranges, and patterns of seasonal breeding of animals.

Biodiversity hot spots- The areas with a high concentration of endemic species facing exceptional levels of threats have been described by Myers in 1988 as biodiversity hotspots. As a whole Sri Lanka has a high degree of endemism. Sri Lanka (wet zone of Sri Lanka) together with Western Ghats of India is considered as one of the Biodiversity hot spot in South Asian region.

Extinction of species

- Existing species have to make room for new species either by changing themselves or by becoming extinct. Therefore natural extinction has to be recognized as part of the evolutionary process.
- The rate of evolution has been generally higher than that of extinction. Therefor there has been an increase in Biodiversity over time.
- Extinction is the elimination of the last member of a species from the earth.
- With the growth of human population and civilization, mankind has greatly increased the rate of extinction.
- The earth today is dominated by humans and no ecosystems on earth's surface is free from human influence.
- In general it is estimated that about 5-10% of the species may face extinction within the next 30 years.

Not only a species but also family or genus or a sub species (taxon) can become extinct if there had been a continuous pressure for its survival. The Red Data book published by the International Union for Conservation of Nature (IUCN) provides a list of threatened species and definie extinct and threatened species as follows.

EXTINCT (EX)

A taxon is Extinct when there is no reasonable doubt that the last individual has died. E.g. Dodo (Lived in Mauritius), Wooly Mammoth (Lived in North America), Legume (*Crudia zeylanica*)

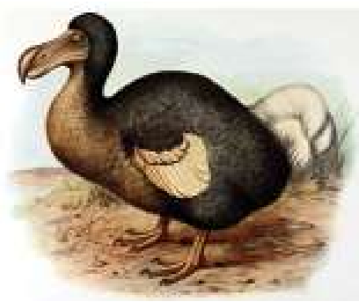

Figure 8.31 : *Dodo*

Extinct in the wild (EW): A taxon is extinct in the wild when it is known only to survive in cultivation, in captivity or as a naturalized population (populations) well outside its natural habitat. E.g; Giant tortoise of Seychelles

Different categories of threatened organisms:

A species is said to be 'threatened' when it is about to leave the world. Threatened organisms are described under three categories. Critically endangered, endangered and vulnerable. The red data book has indicated other categories such as nearly threatened, least concern etc , but these are not considered as a ' threatened species.

CRITICALLY ENDANGERED (CR)

A taxon is critically endangered when the best available evidence indicates that it is facing an extremely high risk of extinction in the wild. Marbled rock frog (E)/Dumbaragalparadiyamadiya (S) and Maha madu (S)/ Ratchatha madupunai (T)- can be listed as one example to represent animals and plants critically endangered in Sri Lanka.

ENDANGERED (EN)

A taxon is endangered when the best available evidence indicates that it is facing a **very high risk** of extinction in the wild. Eg: Etha / Aliya (S)/ Yanai (T), Elephant (E), Wesak Orhid (S/E/T)

VULNERABLE (VU)

A taxon is vulnerable when the best available evidence indicates that it is facing a high risk of extinction in the wild. Punchi Leena (S)/ Dusky-striped, jungle squirrel (E) and Buttercup (E)- are among the vulnerable species found in Sri Lanka .

Endemic species:

An endemic species is a species that is confined to a particular area or country, and not found growing **naturally** anywhere else in the world.

Two examples for plant species endemic to Sri Lanka are:

Dipterocarpus zeylanicus(S: Hora, T: Ennai)

Garcinia quaesita (S: Goraka, T: Gorakappuli)

Examples of two animal species endemic to Sri Lanka are;

Puntius nigrofasciatus (E: Black ruby barb, S: Bulathhapaya, T: Veddiyan)

Loris tardigradus (E: Slender loris, S:Unahapuluwa, T:Thevangu)

Figure 8.32 : Black rubby barb

Indigenous species: A plant or animal species that occurs in its historically known natural range and that forms part of the natural biological diversity of a particular geographic area.

Examples from Sri Lankan indigenous species are:

Lula(S)/ Snake head (E)/ Viral (T)

Kitul(S)/ Thippilipanai (T)

Exotic (alien) species: A species that has been introduced from another geographic region to an area outside its natural range due to human activities. The introduction of species can be intentional or accidental. The following examples represent examples for intentional and direct introductions. Accidental introductions are indirect introductions often considered as 'contaminations' of direct introductions.

Tilapia (S/E), Japan meen (T) for inland fishery industry.

Rubber (S/T), Rupper (T) for plantation industry.

Figure 8.33: Tilapia

Migratory Species: Migration refers to the act of moving from one place to another in a manner that is seasonally determined and predictable. Migration takes place so that organisms can avoid unfavorable environmental conditions that limit breeding.

Suduredi hora(S)/ Indian fly catcher(E) /Inthianeepidipan (T)

Avichchiya (S)/Indian pitta- (E)/ Aarumanikuruvi (T)

Figure 8. 34 : Indian fly catcher

Relict species: The remnants of a once widespread species, which are now found in very restricted or isolated areas, due to fact that areas in which these species are found is lost in many parts of the world.

The Tuatara is an example which lives only on a few small islands of New Zealand. - *Ichthyophis* a primitive ancient legless amphibians with a worm like body. *Lingula* found in Tambalagamuwa bay in Trincomalee is an example for a relict species in Sri Lanka.

Flagship species: Flagship species is a species chosen as a symbol or icon to represent an ecosystem in need for conservation. These species are chosen for their vulnerability, attractiveness or distinctiveness in order to bring about support and acknowledgement from the public at large. Thus, the concept of a flagship species is that the publicity given to few key species, will help the conservation of entire ecosystem and all species contained therein.

The Bengal tiger of India, the giant panda of China, Blue magpie of Sri Lanka- (S: Lanka kehibella, T: Neelavudalperumkuyil) are examples.

Figure 35: Blue magpie

Keystone species: These are species that play a very important role in the stability and functioning of a system. If that species is removed the system tends to collapse. e.g. Planktons of a pond

Invasive alien species- : Invasive alien species are alien (exotic) plants and/or animals whose introduction and spread outside their natural geographic range threaten native biodiversity. Invasive alien species take the advantage of 'human disturbances' in the environment to establish and spread themselves. Their capability to tolerate wide range of environment conditions and high reproductive output help them to easily and successfully expand their populations. Although only a small percentage of alien species become invasive they damage biodiversity (ecosystem, species and genetic levels) in everywhere they invade and alter the services and ecosystem values of the introduced environment. Therefore invasive alien species are considered as a major cause for depletion of biodiversity and environment degradation. The following examples represent invasive alien animal and plant species in Sri Lanka .

Kalutara Golubella (S)/ Giant African land snail (E) was introduced to Sri Lanka as a contamination of soil brought with some other plants. The soil contained eggs of the snail. Japan jabara (S)/ Water Hyacinth (E)/ Kulavazhai (T) was introduced to the country in nearly 110 years ago as an ornamental plant not knowing that it will become a serious invader afterwards.

Conservation

A principal goal of conservation is to ensure the long term survival of as many species as possible. Species that are in danger of extinction have to be specially protected and steps should be taken to ensure their continued reproduction and survival. Conservation can be done in two ways.

In-situ* conservation*: The species is protected and its reproduction facilitated in its natural habitat. Basically a large enough population and adequate, appropriate, habitat space has to be ensured. E.g. National parks such as Yala and Minneriya national parks, Forest reserves such as Kanneliya, Piduruthalagala

Ex-situ* conservation*: The species is taken out of its natural habitats, and looked after in places where its survival and reproduction are ensured.
Zoological gardens and Botanical gardens of a country play a key role in *ex-situ* conservation.

Global Warming and Climate Change

Definition

Climate change refers to a significant, long-term changes of climate attributed directly or indirectly to human activity, that alters the composition of the global atmosphere in addition to natural climate variability observed over comparable time periods (United Nations Framework Convention on Climate Change /UNFCCC, 2011). In the meantime, the Intergovernmental Panel on Climate Change (IPCC) defined the climate change as “a statistically significant change in the state of the properties of the climate which persists for an extended period, typically decades or longer". This definitions refer to any change in climate over time, whether due to natural variability or as a result of human activity.

Global warming is the increase the average temperature of the Earth’s surface (atmospheric and oceanic temperatures) due to enhanced greenhouse effect [or Greenhouse gasses, (GHG)].

Climate change includes warming and the effects arisen due to warming such as melting glaciers, extremes of rains or more frequent drought etc. It can say in another way, global warming is one symptom of much larger problem of climate change.

Contributing factors on global warming and climate change

1. Increase of the emission of CO_2 and other greenhouse gasses (GHG) due to human activities

1.1 Carbon Dioxide (CO_2) is produced by burning of organic matters and it is the most common greenhouse gas to contribute to the global warming and climate change events. Burning of fossil fuel (for running vehicles, electricity generation, industries etc.) is the main cause of the emitting of CO_2 and burning of forests, solid waste are the other main causes contributing the CO_2 emission.

1.2 Methane (CH_4) is another greenhouse gas which has a higher global warming potential. Main emitting sources are anaerobic decomposing (at manure management and waste management), cattle farming, paddy cultivation and enteric fermentation. Even though the CH_4 is a greenhouse gas with high global warming potential, it is less abundant in the atmosphere compared to CO_2.

1.3 Nitrous oxide is (N_2O) another greenhouse gas also with higher global warming potential which is mainly released as by product of fertilizer production and use, other industrial processes, the combustion of certain materials (biomass), nitric acid production, and fossil fuel combustion in internal combustion engines. Nitrous oxide can remain very long time in the atmosphere.

1.4 Manmade industrial gases, namely, Perfluorocarbons (PFCs), Hydrofluorocarbons (HFCs) and Sulfur Hexafluoride (SF_6) are also considered as greenhouse gases with a very high global warming potential.

1.5 Black carbon or the carbon particles which are suspended in lower atmosphere also identified as another cause for global warming. Those particles released as a result of the incomplete combustion of fossil fuels, and other organic matters. These particles are extremely small, ranging from 10 µm 2.5 µm. These Black Carbon have an enormous ability to absorb heart and it cause to increase the air temperature. These particles have ability to absorb heat than CO_2.

Carbon monoxide (CO), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), tropospheric (ground level) ozone (O_3), and nitrogen oxides (NOx) are also considered as gases which have indirect radiative forcing effect. Water vapor, non-methane volatile organic compounds (NMVOCs), and Aerosols are also considered as GHGs.

2. **Deforestation and decrease of the vegetation cover of the world**

   Deforestation is another main cause for global warming. At present the conversion of tropical forest lands to commercial agricultural plantations like palm oil contributes to the mass deforestation all over the world. Forests remove carbon dioxide from the atmosphere and fix through photosynthesis (carbon sequestration). Deforestation reduce the carbon sequestration capacity, carbon capture on the planet (i.e soil carbon) and increase the atmospheric CO_2 concentration.

3. **Destroying the large quantity of phytoplankton by UV radiation due to the deple tion of the ozone layer.**

   This is also similar to deforestation. Phytoplankton which live in warm seas are very useful to maintain the Oxygen and carbon dioxide balance. Normally the carbon absorption capacity of phytoplankton are higher than terrestrial plants. Though the phytoplankton are unicellular microscopic organism they have been distributed in large area and they were responsible for the 60%-70% of absorption of atmospheric carbon. Due to depletion of the ozone layer UV radiation which comes from sun can destroy this kind of tiny organisms and may cause to reduce the CO_2 absorption capacity of oceans and increase the global temperature.

Effects of the global warming and climate change

1. **Sea level rising**

   Scientists predict an increase in sea levels worldwide due to the melting of two massive ice sheets in polar regions and thermal expansion of water. However, many island nations around the world will experience the effects of sea level rise.

2. **Extreme weather events**

   Occurrence of extreme weather events such as prolong drought, intensive rainfall and its effects such as floods and landslides and storms have been increased for last decades was considered as a negative impact of climate change. It causes loss, damages and disasters.

3. **Less production of foods (Threat to food security)**

   Due to unexpected extreme weather condition the crop production will be getting loss around the world. It may be due to heavy rains or the severe droughts.

4. **Degradation of coral reefs**

   The corals bleaching and degradation due to rises in sea temperature is a severe danger for whole ocean ecosystems and many other species which lives on coral reefs for their survival. According to the recent reports coral populations will collapse by 2100 due to increased temperatures and ocean acidification and its effects

5. **Increase the insect population.**
   Due to increase of the insect population some diseases such as mosquito-borne malaria and dengue will spread than present. As well the over growth of the insect populations will be a massive threat to food production

6. **Loss of Biodiversity**
   Climate change and global warming may cause reduced biodiversity, Ecosystems structure will be change and some species may move from their present range of occurrence and may survive while some others will not be able to move and could become extinct.

Ozone Layer depletion

Most ozone particles are concentrated in the region of stratosphere in between 10-50 km of the atmosphere, and this layer is identified as the 'Ozone Layer'. This is very much important to protect living things from burning from too much ultraviolet (UV) radiation emitting from the sun

Naturally, the total concentration of ozone in the stratosphere remains relatively constant with the concentration of 300 to 350 Dobson Units (D.U). Ozone depletion is described when levels fall below 200 D.U due to man-made ozone depleting chemicals (ODCs) as in the stratosphere over the South Polar Region. This thinning of the ozone layer refers as Ozone hole.

Contribution factors for the Ozone layer depletion

Ozone depletion occurs when the natural balance between the production and destruction of stratospheric ozone is disturbed or lost. This is mainly due to chlorine and bromine released from man-made compounds called ozone depleting substance such as CFCs, MeBr, Helene and HCFC.

Effects of the Ozone Layer depletion

Depletion of the ozone layer in the stratosphere will increase penetration of solar UV-B radiation which is likely to have profound impact on human health with potential risks of eye diseases, skin cancer and infectious diseases.

It is a known fact that the physiological and developmental processes of plants are affected by UV-B radiation.

UV-B radiation is likely to result in changes in species composition (mutation) thus altering the bio-diversity in different ecosystems. It directly destroyed phytoplankton which form the foundation of aquatic food webs in the sea and it cause to reduce the composition of food web in the sea ecosystem.
UV-B can also cause damage to early development stages of fish, shrimp, crab, amphibians and other animals.

Desertification

Desertification means “process of land degradation in arid, semi-arid and dry sub-humid areas resulting from various factors, including climatic variations and human activities.”(UNCCD, Paris, 1994),

Contribution factors for desertification

The main driving forces of desertification can be separated into climate variations and human activities, according to the UNCCD definition. It is known that human activities have a great influence in the climate change and it has caused desertification

Deforestation is another main factor to decertification because of it is directly influencing to reduce the rainfall, precipitation, soil humidity and water recharge of underground reservoirs

Over-exploitation of water and soil, uncontrolled mining and excessive use of agro-chemical products and as well as poor land management practices also caused to desertification.

Effects of the Desertification

Decrease ecosystems services and reduced biodiversity in affected areas. Decreasing of the vegetation cover induce water scarcity, destroys habitats of animal and plant species and reduces agricultural activities mainly the growth of crop species. It may effect to the food security of the people as well as for the animals. The process of desertification presents a serious impact on the human well-being and health of the people living in the areas affected by droughts and land degradation

Due to the desertification carbon storage capacity of plants and soils will also be reduced in the long run.

Acid rains

Acid rain is one of the most serious global environmental issue which, emerged due to air pollution mainly due to Sulphur dioxide (SO_2) and oxides of nitrogen. These pollutants are released from anthropogenic activities such as combustion of solid waste, fossil fuels in thermal power plants and vehicle engines.

Acid rain is a commonly used term for acid deposition, which includes rain, snow, fog and dry particles that fall from the sky. Uncontaminated precipitation is also slightly acidic and normal rain water may be about a 5.6 on the pH scale because carbon dioxide is dissolved and carbonic acid is formed. Acid rain may show a lesser pH.

Contributing Factors of Acid Rains

Mainly caused by the release of sulfur dioxide and nitrogen oxides into the atmosphere due to burning of fossil fuels, where they are converted into sulfuric and nitric acid respectively.

Effects of the Acid Rains

Acid rain directly causes severe damage to building structures and marble statues

Increases the acidity of fresh water ecosystems such as streams, lakes, and marshes and it causes to change the structure and the composition of above fresh eater ecosystems.

Cause to destroy soil organisms and loss the soil fertility.

It cause to leach metals such as copper, aluminum, and some heavy metals such as lead and mercury in the soil and into runoff as well as drinking water

Acid rain causes significant damage to forests. It directly affects trees and other small plants which are important to the ecosystem

The Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES)-1975

The aim of CITES is to ensure that international trade of specimens such as horns and skins of wild animals and whole plant or parts of plants does not threaten their survival. According to CITES, the export of some species (as listed) requires prior permission and presentation of an export permit. An export permit shall be granted only if export of that species will not be detrimental to that species survival. Some examples from Sri Lanka are skin of Sri Lankan Leopard (S: Kotiya, T: Siruththaipuli), plants of *Cycas* - Madu (S) / Madupanai (E).

Conventions related to conservation of the environment

Convention on Biological Diversity (CBD)- 1992

The Convention on Biological Diversity (CBD), known informally as the Biodiversity Convention, addresses all aspects of Biodiversity conservation.

It has three main goals:

- the conservation of biological diversity (or biodiversity)- e.g. conservation of genetic materials and species and ecosystems.
- the sustainable use of components of biological diversity- e.g. imposing limits to control overexploitation.

- The fair and equitable sharing of benefits arising from genetic resources- e.g. exchange of genetic material, species between countries without conflict.

The Convention on Wetlands (Ramsar Convention)-1971

The Convention on Wetlands, called the Ramsar Convention, provides the framework for the conservation and wise use of wetlands and their resources. In Sri Lanka, there are six wetlands declared as Ramsar sites: Anawilundawa, Bundala, Kumana, Maduganga, Vankalai and Wilpattu.

Figure 8.36 : Anawilundawa wetland

International Convention for the Prevention of Pollution from Ships (MARPOL)- 1973

International Convention for the Prevention of Pollution from Ships (MARPOL) is the main international convention covering prevention of pollution of the marine environment by ships from operational or accidental causes. The objective of this convention is to preserve the marine environment in an attempt to completely eliminate pollution by oil and other harmful substances and to minimize accidental spillage of such substances.

Montreal Protocol- 1989

Montreal Protocol on Substances that Deplete the Ozone Layer is an international treaty designed to protect the ozone layer by phasing out the production of numerous substances that are responsible for ozone depletion.

Kyoto Protocol-1977

The Kyoto Protocol is an international agreement linked to the United Nations Framework Convention on Climate Change (UNFCCC), which commits its Parties by setting internationally binding green house gases (GHGs) emission reduction targets. In 2012 Doha, Qatar, the "Doha Amendment to the Kyoto Protocol" was adopted. During this commitment period (2013-2020) parties are committed to reduce GHG emissions by at least 18 percent.

Basel Convention -1989

The Basel Convention on the Control of trans boundary movements of hazardous wastes and their Disposal intends to protect human health and the environment against the adverse effects of hazardous wastes. Its scope of application covers a wide range of wastes defined as “hazardous wastes” based on their origin and/or composition and their characteristics, as well as two types of wastes defined as “other wastes” such as waste containing heavy metals such as Lead (Pb) and Mercury (Hg) and wastes from hospitals which contains contaminants.

Environmental Legislation and Policies in Sri Lanka

The Constitution of the Democratic Socialist Republic of Sri Lanka has provision for the protection, preservation and improvement of the environment. The government of Sri Lanka has formulated many legislation and policies focused on environmental conservation and most of them are revised from time to time to accommodate updates. Legislation is a set of regulations for which stakeholders is given a penalty when it is violated. Policy is a set of practices that is followed by stakeholders and there is no such penalty imposed when it is not practiced.

Fauna and Flora Protection Ordinance and National Environmental Act (NEA) are Some examples for key legislation on environment conservation.

Fauna and Flora Protection Ordinance (FFPO)

The Ordinance, No.2 of 1937, and subsequent amendments, make provisions for the protection of wildlife and flora in the country. The authority responsible for enforcing this law is the Department of Wildlife Conservation (DWLC). Operation and management of National Parks, Strict Natural Reserves, Jungle Corridors and Sanctuaries are conducted under FFPO.

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Notes:

This is to acknowledge that some of the diagrams used in this book have been taken from various electronic sources using internet . This book is not published to make profit and sold only to cover cost.

The resource book is prepared according to the subject content and learning outcomes of the G.C.E. (A.L) Biology new syllabus which is implemented from 2017.

The content of this Resource book declares the limitation of the G.C.E. (A.L) Biology new syllabus which is implemented from 2017.

09

Microbiology

Nature of microorganisms

Microbiology is the study of organisms that are too small and are not visible clearly to the naked eye or un-aided eye when they exist individually. These organisms are referred to as microorganisms. Microorganisms include bacteria, archaea, cyanobacteria/ Blue green bacteria (BGB), fungi and protists. Mollicutes such as mycoplasmas and phytoplasmas, viruses, viroids and prions are also studied under microbiology.

Microscopic nature of microorganisms

In general, microorganisms are less than 0.1 mm in size and cannot be observed with unaided eye. Therefore, they must be observed with a microscope. Microorganisms and their structural components are measured in micrometers and nanometers.

1 micrometer (µm) = 10^{-6} m

1 nanometer (nm) = 10^{-9} m

Some microorganisms are more readily visible than other because of their larger size.

Ubiquitous nature of microorganisms:

Microorganisms are ubiquitous on earth. They are found in water, soil, air and interior and exterior surfaces of other organisms. Marine and freshwater microorganisms form the basis of food chain in oceans and freshwaters. Some of them do photosynthesis and are primary producers in aquatic environments. Soil microorganisms help recycling of chemical elements between soil, water, air and living organisms. Microorganisms suspended in air as bioaerosols, have the opportunity to travel long distances with the wind current and precipitate. Pathogenic bioaerosols, cause opportunities for disease spreading. Only a minority of microorganisms that associate with other organisms such as plants, animals and human are pathogenic. Majority of them are advantageous or harmless. However, all viruses are harmful to the organisms they attached to. Some microorganisms are capable of inhabiting extreme environmental conditions that are unfavorable or even lethal for other organisms. Such microorganisms are known as extremophiles. Extremophiles have been found inside the Earth's crust, deep sea at high pressures, extreme acidic or extreme basic conditions, hydrothermal vents, frozen sea water and anaerobic conditions. Extremophiles are classified according to the conditions in which they grow.

Table 9.1: Types of extremophiles

Type of Extremophile	Condition
Thermophiles	high temperatures
Psychrophiles	low temperatures
Acidophiles	acid pH
Alkaliphiles	basic pH
Halophiles	require NaCl
Barophies	high pressure

Some of these extreme environments consist of more than one extreme condition. For example,

- Many hot springs are acidic or alkaline in nature at the same time
- Deep seas are cold and remain in high pressure.

Microorganisms live in such environments are adapted to live with more than one extreme condition.

High growth rate of microorganisms:

Rates of growth and reproduction of microorganisms are high. Microorganisms possess a high surface area/volume ratio due to their smaller size. This means that they have large surface area available for exchange of materials from external environment. As a result, flowing rate of materials in to the inside of cells and the exit of waste materials to the outside of the cells increases and results in high metabolic rate. Therefore, average generation time or the time required to double the population of microorganisms is relatively less.

Morphological, nutritional, and physiological diversity of microorganisms:

Microorganisms possess diverse morphological forms. Bacteria possess diversity in their shapes, basically three distinct shapes; rod shape/ bacillus, spherical shape/ coccus and spiral shape/ spirillum. The coccus bacteria may arranged in different forms; coccus/ monococcus, diplococcus, streptococcus, staphylococcus, tetrads and sarcinae. Bacillus bacteria may arranged in to either diplobacillus or streptobacillus. Spiral bacteria may arranged in to either vibrio or spirillum or spirochete.

Cyanobacteria exhibit a great variety of shapes and arrangements, unicellular to long multi cellular filaments. Multi cellular Cyanobacteria may appear as either filamentous or non-filamentous. Filamentous appear as chains and the non-filamentous appear as clusters or colonies forming spherical, cubical, square or irregular shape. Two morphological varieties are found in viruses based on their symmetry of protein coats; Icosahedral and helical. In fungi, some of them are unicellular and others multicellular, consists of a mass of fine tubular branching threads known as hyphae, collectively form mycelium. Hyphae may be septate or aseptate. Prions are smaller proteinaceous particles. Unicellular protists possess wide range of morphological diversity. Mollecutes are pleomorphic (variable shapes).

Microorganisms show a diversity of nutritional types. Based on the sources of carbon and energy, nutritional types of microbes are classified. There are four major nutritional types seen among microorganisms; chemoautotrophs, chemoheterotrophs, photoautotrophs and photoheterotrophs. Based on the utilization of $O_2(g)$, microorganisms can be classified in to four physiological groups; obligate aerobes, obligate anaerobes, facultative anaerobes and microaerophiles. Some microbes capable of fixing atmospheric molecular nitrogen, show physiological diversity; free-living nitrogen fixing microbes and symbiotic nitrogen fixing microbes.

Types of microorganisms

1. Bacteria

Bacteria (singular, bacterium) are single-celled (unicellular) prokaryotic organisms. They show different morphological forms and arrangements. The most obvious structural feature of bacteria is the shape of individual cells. There are three basic shapes.

1. Spherical; coccus (plural- cocci)

2. Rod shape; bacillus (plural-bacilli)

3. Spiral shape; (plural- spirilli)

During cell division, cells can remain attached to each other and form different forms of cell arrangements.

1. Different forms of cell arrangement of coccus bacteria (Figure 9.1)

Coccus	Cells divide in one plane. Divided cells detach from each other after cell division
Diplococcus	Cells divide in one plane. Divided cells remain in pairs
Streptococcus	Cells divide in one plane. Divided cells remain attached in chain like pattern
Tetrad	Cells divide in two planes producing 4 cells remain attached together
Sarcinae	Cells divide in three planes and remain attached in groups of 8 cells
Staphylococcus	Cells divide in multiple planes and form grape like clusters of cells

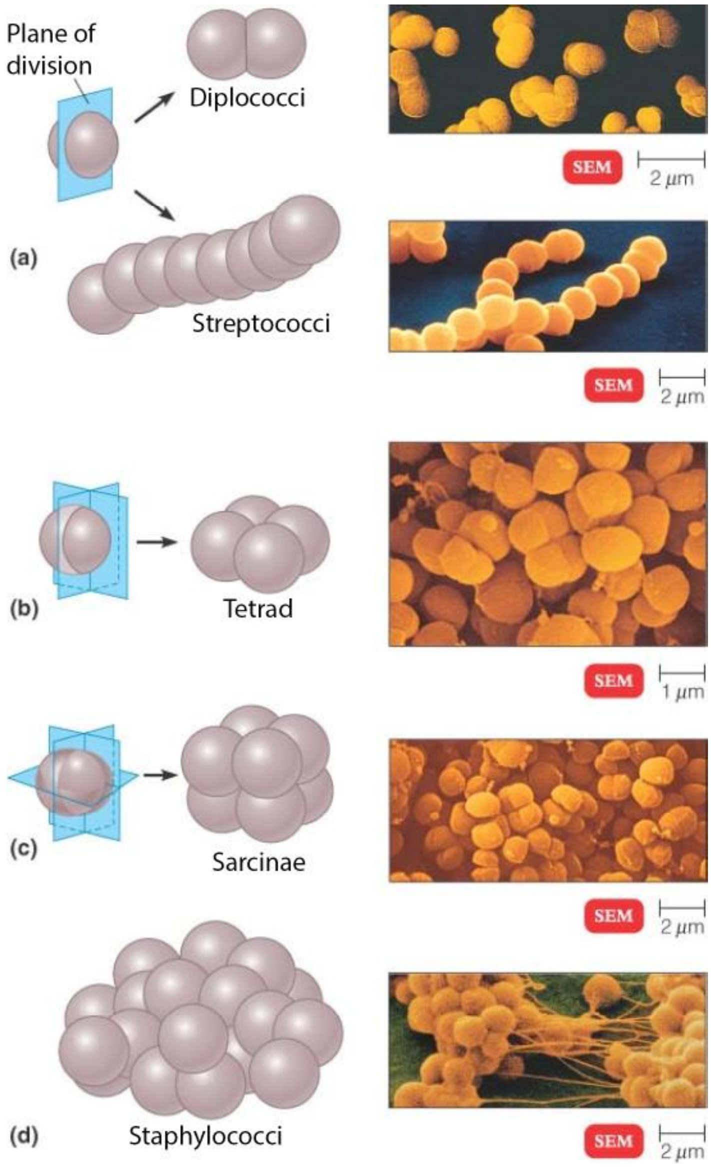

Figure 9.1. Cell arrangement of coccus bacteria. Shown are diagrammatic representations of planes of cell division and various cell arrangements (left) and their scanning electron microscopic (SEM) views (right). Retrieved from Microbiology: An Introduction, 11th edition, by Tortora, Gerard J.; Funke, Berdell R.; Case, Christine L. Pearson Education Ltd (2016).

2. Different forms of cell arrangement of bacillus bacteria (Figure 9.2)

Bacilli divide only across their short axis. Therefore, there are few cell arrangement forms.

Single bacillus	Single rod
Diplobacillus	Remain in pairs after cell division
Streptobacillus	Occur in chains after cell division

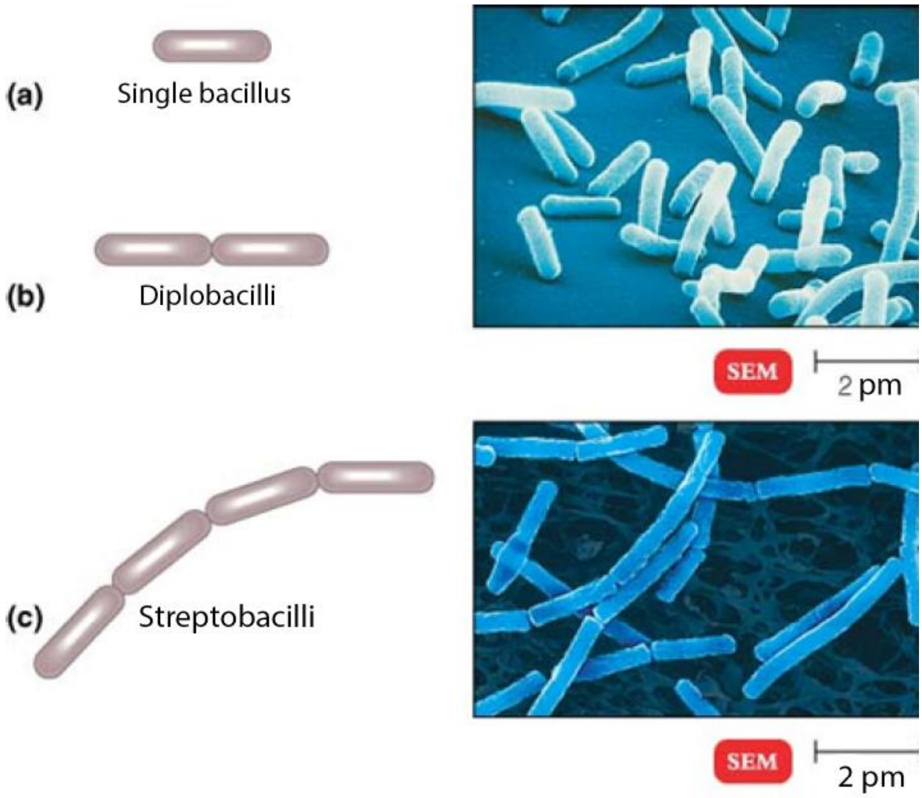

Figure 9.2. Cell arrangement of bacillus bacteria. Shown are diagrammatic representation of (left) and scanning electron microscopic (SEM) views of cells (right). Retrieved from Microbiology: An Introduction, 11th edition, by Tortora, Gerard J.; Funke, Berdell R.; Case, Christine L. Pearson Education Ltd (2016).

3. Different forms of cell arrangement of spiral bacteria (Figure 3)

Spiral bacteria have one or more twists, they are never straight.

Vibrio	Bacteria look like curved rods
Spirillum	Helical shape, like a corkscrew and rigid body
Spirochete	Helical shape, flexible body

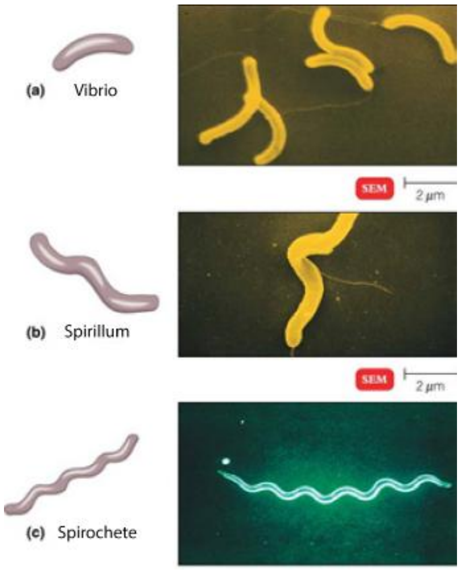

Figure 9.3: Shapes of spiral bacteria. Shown are diagrammatic representation of shapes (left) and scanning electron microscopic (SEM) views of spiral bacteria (right). Retrieved from Microbiology: An Introduction, 11th edition, by Tortora, Gerard J.; Funke, Berdell R.; Case, Christine L. Pearson Education Ltd (2016).

Bacteria show a diversity of nutritional types. Four major nutritional types can be identified among bacteria. They are classified based on the source of energy and carbon.

Nutritional type	Source of energy	Source of carbon	Example
Photoautotrophs	Light	Carbon dioxide (Inorganic Carbon)	Purple sulfur and Green sulfur bacteria
Photoheterotrophs	Light	Organic carbon	Purple non sulfur bacteria
Chemoautotrophs	Inorganic chemicals	Carbon dioxide (Inorganic Carbon)	*Nitrobacter, Nitrosomonas, Thiobacillus thiooxidans*
Chemoheterotrophs	Organic chemicals	Organic carbon	Most bacteria

Microorganisms can be classified into 4 groups on the basis of their tolerance to oxygen.

Physiological Group	Description	Example
Obligate aerobic	These bacteria require oxygen for their survival. They generate energy by oxidative phosphorylation.	*Acetobacter sp*
Obligate anaerobic	They cannot survive in the presence of oxygen. These microorganism generate energy by fermentation	*Clostridium* sp.

Facultative anaerobic	These microorganisms prefer to grow in the presence of oxygen producing energy by oxidative phosphorylation, but they can also grow in anaerobic environments using fermentation	*Escherichia coli*
Microaerophilic	These microorganisms can grow only in oxygen concentrations lower than those in air	*Lactobacillus* sp.

Some bacteria are able to fix atmospheric nitrogen. They show diversity in nitrogen fixation.

- Free-living nitrogen fixing bacteria: *Azotobacter* sp.
- Symbiotic nitrogen fixing bacteria: *Rhizobium* sp. with legume root

Mostly bacteria undergo asexual reproduction by binary fission, and in some occasion, fragmentation or budding. In rare occasions, bacteria of two strains share a portion of genetic material through the process of 'conjugation'.

Cyanobacteria

Cyanobacteria are named for their characteristic blue-green (cyan) pigmentation. Cyanobacteria also exhibit a great variety of shapes and cell arrangements, unicellular to colonial forms (Figure 9.4).

- **Unicellular form-** Cells separate after cell division. However, in nature majority of unicellular forms stay together by copious secretion of mucilage by daughter cells.
- **Colonial form-** Cells remain attached by walls or held in a common gelatinous matrix forming a colony of cells. Colonies may either be non-filamentous or filamentous. Non-filamentous colonial form- depending on the plane of division and direction there are different arrangements such as spherical, cubical, square or irregular shape. Filamentous colonial form is the result of cell division in a single plane and a single direction forming a chain or thread like structure.

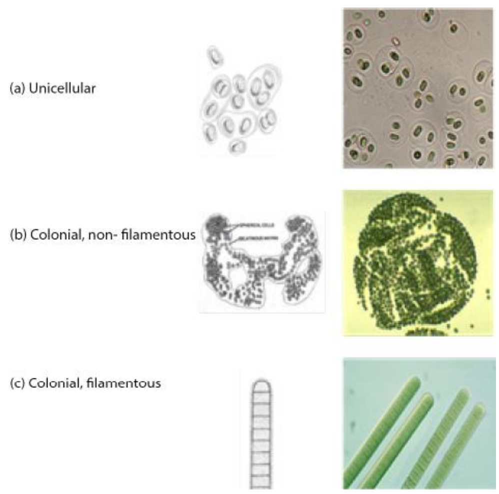

Figure 9.4. Cell arrangement of cyanobacteria. Shown are diagrammatic representations (left) and light microscopic views (left) of cell arrangement.

Cyanobacteria are photoautotrophs that carryout oxygenic photosynthesis similar to plants and algae. Many of cyanobacteria are capable of fixing atmospheric nitrogen. For examples, *Nostoc* sp. is a free living nitrogen fixing cyanobacteria, *Anabaena -Azolla* symbiotically fix nitrogen with its partner, *Azolla* sp. (water fern). In most cases nitrogen fixation takes place in special cells called heterocyst. Nitrogen fixation is catalyzed by the enzyme called nitrogenase in the heterocyst. Nitrogenase is sensitive to oxygen. Heterocyst carry thick cell wall to protect nitrogenase from oxygen that could diffuse from neighboring photosynthetic cells and from air or water.

Cyanobacteria carry another specialized cell type called akinete. They are thick walled resting spores with stored food. Akinete is resistant to drought and high temperature. Therefore, akinete is able to survive during unfavorable environmental conditions although vegetative cells dries out.

Cyanobacteria reproduce only by asexual methods. Single unicellular and colonial non-filamentous types undergo simple cell division while colonial filamentous and colonial unicellular forms reproduce by fragmentation.

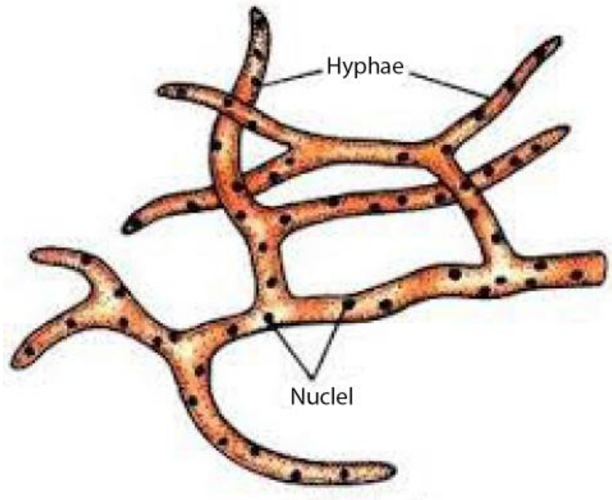

figure 9.5 Fungi fillamentous thallus, branched mycellium

Fungi (singular, fungus) are eukaryotes. They may be unicellular (yeast) or multicellular (molds). Some multicellular fungi form mushrooms. Molds form visible masses called mycelia, which are composed of long filament like structures called hyphae. Many molds contain cross-walls called septa (singular, septum). Septa divide hyphae into distinct single nucleate cell-like units. Some molds do not contain septa in their hyphae resulting in long continuous cells with many nuclei. These are called coenocytic hyphae. The cottony growths sometimes found on bread and fruit are mycelia of molds.

Fungi are chemoheterotrophs and acquire food by absorption. They possess saprophytic mode of nutrition. They play important role in food chain by decomposing dead plant materials by secreting enzymes and thereby recycle vital elements. Parasitic (plant and animal pathogens) and mutualistic (lichens and mycorrhizae) modes of nutrition also found among fungi.

Unicellular fungi reproduce asexually by fission or budding, on the other hand, filamentous fungi (molds) reproduce asexually and/or sexually by producing spores.

a.

Figure 9.6 (a). Reproduction method of *Penicillium (*spore formation)

b.

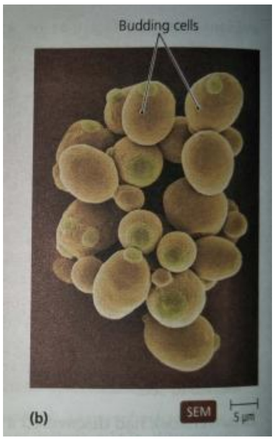

Figure 9.6 b. Reproduction method of Yeast (budding)

Unicellular protists

a.

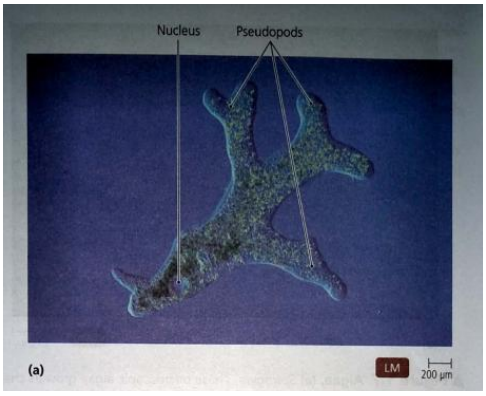

Figure 9. 7 a. *Amoeba*

b.

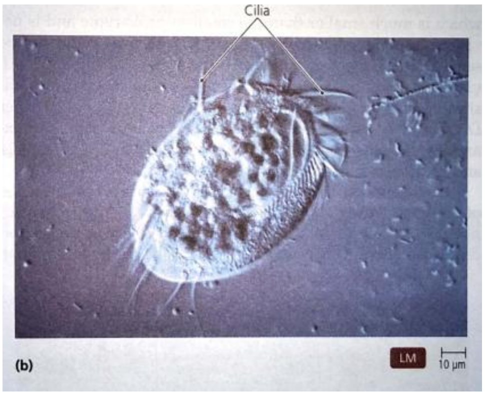

Figure 9. 7 b. *Paramecium*

c.

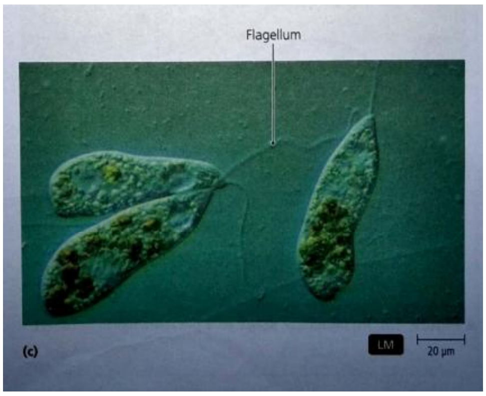

Figure 9. 7 c. *Euglena*

Unicellular Protists are pleomorphic, vary in their shapes and possess locomotive structures such as pseudopods, cilia or flagella. They exist either individually or form colonies. Some join together and form filaments.

Photoautotrophic, heterotrophic and mixotrophic modes of nutrition are found among protists.

There are aerobic, anaerobic and facultative anaerobic respiratory modes found among protists.

Some algae contribute to the symbiotic interactions in lichens.

They reproduce sexually by gametes and asexually by fission.

Mollicutes

Mollicutes are prokaryotes included in the domain bacteria. Mycoplasma and phytoplasma, are considered unique due to absence of cell walls.

Mycoplasma and Phytoplasma

Mycoplasma are pleomorphic, vary in shape from spherical to filamentous. They are the smallest prokaryotic group of organisms invisible under light microscope. Mycoplasma do not contain flagella. Almost all mycoplasma are parasites of humans and animals. Mycoplasma require high amount of organic growth factors. They reproduce by budding and binary fission and do not produce spores. Mycoplasma are aerobic or facultative anaerobic.

Phytoplasma resembles to mycoplasma in many ways. They are similar in size to mycoplasma. Both can only be seen under electron microscope. Shape varies from spherical to filamentous. Phytoplasma only infect plants and are generally present in the phloem sap. They cannot grow in artificial media. They are transmitted mostly by leafhoppers. Therefore, they reproduce in both leafhoppers as well as plant body. They reproduce by budding and binary fission. They possess aerobic or facultative anaerobic mode of respiration.

Virus

(a) Characteristic features

Viruses are neither prokaryotes nor eukaryotes and do not show any cellular organization. They do not possess any metabolic activity or reproduction when they are out of living host cells. Thus, they are not considered as living organisms. However, once they get in to the host cells, they multiply and cause infection through various metabolic pathways, shows characteristics of living organisms. Since viruses can only multiply within a living host cell, they are obligate parasites. Viruses are very small can only be seen through an electron microscope. Viruses possess simple structures, usually are composed of a central core of a nucleic acid and surrounded by a protein coat called the capsid which is made up of a fixed number of protein subunits called capsomeres. Viruses may have either DNA or RNA as their genetic material. They do not have protein synthesis machinery such as additional RNAs or enzymes for protein synthesis. Therefore, they depend on host cell's protein synthesis machinery. RNA viruses consist of reverse transcriptase enzymes for reverse transcribing RNA in to DNA.

(b) Morphology and types of viruses

On the basis of capsid architecture two basic morphological symmetries can be identified (Figure 9.8).

1. Helical

2. icosahedron

Based on the above symmetries viruses show four types of morphological forms; helical, polyhedron, complex and enveloped.

- Helical viruses- long rigid or flexible rods. e.g; Rabies virus
- Icosahedron/ polyhedral- icosahedron symmetry. e.g; Adeno virus
- Complex viruses-Exhibits more than one form of symmetry with additional structures. e.g; bacteriophage
- Enveloped viruses. e.g; roughly spherical. Capsid covered by envelopes. e.g; Herpes simplex virus

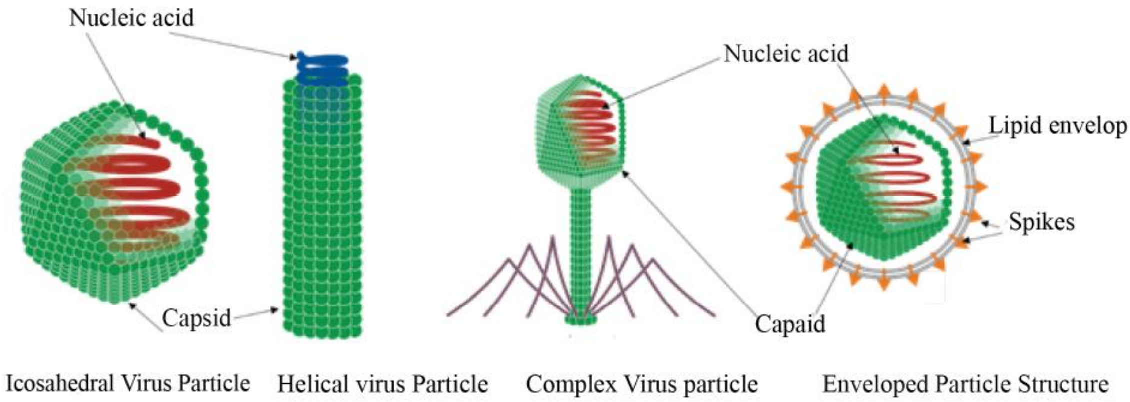

Figure 9.8 Diagrammatic representation of four morphological forms of virus.

Multiplication of viruses

A single virus can give rise to thousands of similar viruses in a single host cell. Therefore, viruses cause serious damages to their host leading to severe diseases in plants, animals and bacteria. Bacteriophages are typical group of viruses that are capable of infecting bacteria. They multiply by two distinct mechanisms; lytic cycle or lysogenic cycle.

Lytic cycle involves with the lysis of the host cell, whereas the lysogenic cycle allows viral DNA incorporating into host DNA and multiply without causing lysis of the host cell.

Lytic cycle of a bacteriophage

There are five distinct steps in the lytic cycle ; attachment, penetration, biosynthesis, maturation and release (Figure 9.9).

- **Attachment:** The first step is the attachment of virus to a matching receptor site on the bacterial cell.
- **Penetration**: After attachment, bacteriophage injects its DNA into the bacterial cell. This is facilitated by an enzyme which breaks down bacterial cell wall.
- **Biosynthesis:** The next step is the biosynthesis of viral DNA and proteins in the host cytoplasm using host resources. This stage induces degradation of host cell DNA.
- **Maturation and assembly:** Once bacteriophage DNA and proteins are synthesized, DNA and capsid are assembled to form complete virus particles. This is called maturation.
- **Release:** Finally, bacteriophage induce bacterial cell to break open (lyse). Newly produced bacteriophages are released from the host cell. These released bacteriophages can start another lytic cycle in cells in the vicinity.

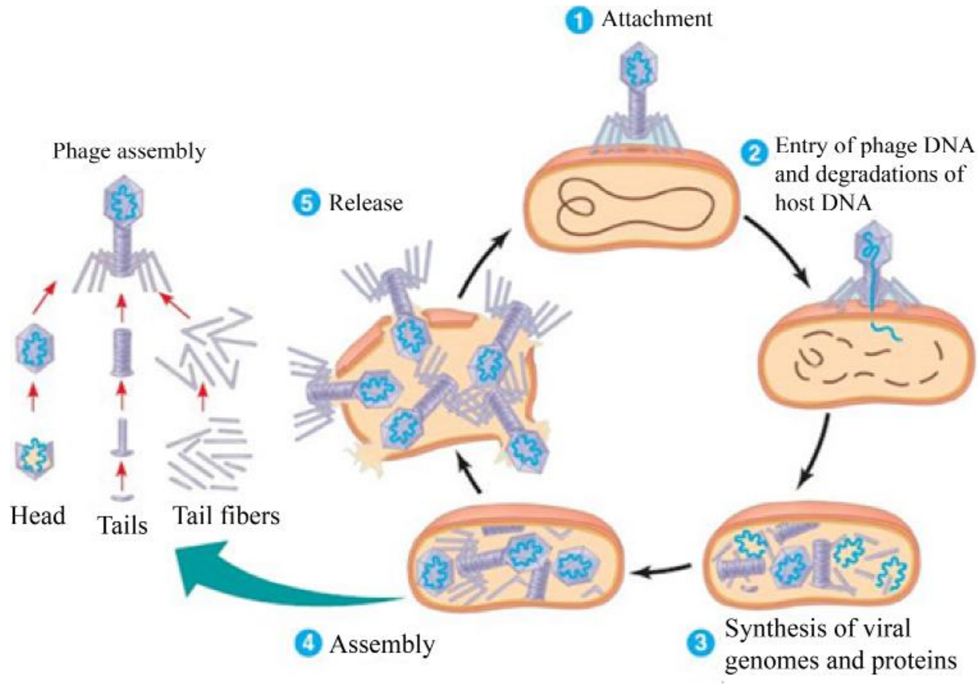

Figure 9.9 Stages of lytic cycle of a bacteriophage

Viroids

Viroids consist only of short piece of naked RNA with no protective layer such as a protein coat. Viroids can only multiply within a living host cell using host cell resources. However, viroids do not contain any gene and only carry signals for their multiplication. Viroids infect plants, but no other life forms till to date.

Prions

Prions are proteinaceous infectious particles. Their size is smaller than virus. Although prions lack nucleic acid they can replicate with the help of a host's gene that encodes the prion protein. They are found as disease causing agents in some birds and mammals. All these diseases are neurological diseases. Eg;

- Transmissible Spongiform Encephalopathies (TSEs), because large vacuoles develop in the brain giving sponge-like appearance.
- Mad cow disease was one of the serious disease emerged in cattle in 1987.
- Creutzfeldt-Jakob disease (CJD) is one of the human diseases caused by prions.

Human to human disease transmission has been associated with transfusion of infected blood and tissue and organ transplantation. Some TSE infections may be transmitted from cow to human.

Basic laboratory Techniques

For the study of morphology and biochemical properties of microorganisms, it is essential to culture them on artificial media. There are some basic laboratory techniques such as preparation of artificial culture media and sterilization techniques, to be followed in culturing of the microorganism of interest without any contamination. This section describes such basic laboratory techniques.

Methods of sterilization

Sterilization is the process of removal or destruction of all forms of microbial life including endospores.

There are two types of sterilization, physical and chemical.

I. Physical methods of sterilization

Sterilization by moist heat, dry heat, filtering using membrane filters, exposure to UV radiation are some of the physical methods used in sterilization.

- **Moist heat sterilization**

  Here, moist heat is used to destroy the microorganisms present in the desired materials such as culture media, temperature labile reagents/ fluids and various laboratory utensils. This is done by denaturing of proteins by high temperature and pressure.

E.g. autoclaving- In an autoclave, steam with 121 ºC temperature at the pressure of 1 atm/ 15 psi is used for sterilization. Extending the above condition for 15 minutes is sufficient to kill all microorganisms (except prions) and their endospores.

Autoclaving is used to sterilize culture media, solutions, syringes and needles, healthcare instruments and various other items that can withstand high temperatures and pressure. Glassware can also be sterilized with an autoclave if care is taken to ensure that the steam contacts all surfaces.

Pressure cooker also can be used for moist heat sterilization.

- **Dry heat sterilization**

  Here, dry heat is used to destroy the microorganisms present in the desired materials such as glasswares, petridishes, pipettes, inoculation loops, inoculation needles, scalpels, etc.

1. **Direct flaming**

   It is a simple method of dry heat sterilization. This is used in laboratories to sterilize inoculating loops, inoculating needles and scalpel blades by heating them on the flames of Bunsen burners / hot spirit lamp until they reach red hot.

2. **Incineration**

   It is mostly done in an incineration oven. Incineration is used to sterilize hospital waste. Microorganisms are burned to ash during direct flaming and incineration.

3. **Hot-air sterilization**

   Microorganism are killed by oxidation. Items to be sterilized are heated to about 170 ºC and maintain for 2 hours in a dry air oven. This type of sterilization is often used to sterilize glassware such as Petri plates, flasks, beakers, bottles and glass pipettes.

- **Pasteurization**

  Louis Pasteur found spoilage of beer and wine can be prevented by applying mild heat that kills organisms causing spoilage without seriously damaging the taste, texture and nutritional content of the product. Later, the same principle was applied to milk products, now known as pasteurized milk. The objectives of milk pasteurization are to eliminate pathogenic microorganisms and reduce microbial number which prolongs milk quality under refrigeration.

  High temperature short-time (HTST) pasteurization, which uses temperature of at least 72 ºC for 15 seconds and **low temperature long time (LTLT)** 63 ºC for 30 minutes are the two main methods of pasteurization. Milk can also be sterilized by **ultra-high-temperature (UHT)** pasteurization. Here, milk is heated to about 140 ºC in less than 5 seconds by flashing steam. This milk can be stored for several months without refrigeration.

- **Boiling**

  Boiling the materials such as surgical instruments to 100 ºC. Most of the pathogenic microorganisms are killed at boiling temperature.

- **Filtration- e.g. Membrane filters**

  Filtration is used to sterilize heat sensitive liquids such as solutions containing enzymes, vitamins, antibiotics, vaccines and some culture media. Material to be sterilized is passed through a filter by using vacuum. Filter retains microorganism and the liquid is passed through the filter.

  Membrane filters are widely used to sterilize heat sensitive solutions. Pores of membrane filters are from 0.01 µm to 0.45 µm size, retain almost all the microorganisms including viruses and some large protein molecules.

- **UV Radiation**

  UV Radiation kills microorganisms that falls in to direct exposure, either through destruction or damaging DNA. However, a major disadvantage of UV is that radiation does not penetrate through solid surfaces and coverings such as paper, glass and textile. Therefore, anything to be sterilized should have direct contact with radiation. UV radiation is commonly used to sterilize air in hospital rooms such as operating theaters and nurseries.

II. Chemical methods of sterilization

Few chemicals such as ethylene oxide and chlorine dioxide (both are gases) are currently used as chemical sterilizing agents. Majority of chemical agents reduce microbial populations to a safe levels or remove vegetative forms of pathogens.

Ethylene oxide kills microorganisms and endospores. It is also highly penetrating. Therefore ethylene oxide is used to sterilize mattresses in hospitals.

Chlorine dioxide has been used to fumigate enclosed building areas contaminated with endospores of Bacillus anthracis. It is most commonly used in water treatment prior to chlorination.

Preparation of culture media

Microorganisms cannot be studied in their natural habitat such as in soil, water or air. Therefore, we need to bring them to the laboratory and provide similar conditions for their growth and reproduction. A nutrient material prepared for providing nutrition and anchorage essential to the growth of microorganisms at laboratory condition is called a culture medium.

Not all microorganism can be grown on a laboratory culture media. They are called non-culturable microorganisms. Some microorganisms grow well on any culture medium whereas other microorganisms require special medium.

Suppose we want to grow a culture of microorganisms present in a certain soil sample. The culture medium should contain necessary nutrients, sufficient moisture and suitable pH. This medium must initially be sterile which means it should not contain any living microorganisms. Therefore, when preparing a culture medium all glassware and liquid nutrient solutions should be sterilized.

Nutrient agar (NA) and potato dextrose agar (PDA) are two general media, commonly used to grow bacteria and fungi respectively. Nutrient agar is made up of peptone, meat extract, sodium chloride, agar and distilled water. Whereas, PDA is made up of potato, glucose, agar and distilled water.

Here, agar is used as a solidifying agent. Agar solidifies at temperatures below 40 °C which means a culture medium containing agar is a solid medium. For growing microorganisms, solid culture medium is usually contained in Petri dishes or test tubes.

Because most microorganisms appear almost colorless when viewed through a standard light microscope, we have to prepare them for observations. One of the ways this can be done is by staining, means coloring the microorganisms with a dye.

However, before the microorganisms stain, they must be fixed (attached) to the microscopic slide. A simple stain is an aqueous or alcohol solution of a single basic dye. The primary purpose of a simple stain is to highlight the entire microorganisms so that cellular shapes, cell arrangements and basic structures are visible. Some of the simple stains commonly used in the laboratory are methylene blue., crystal violet and safranin.

Microorganisms and diseases

Generally, humans are free of microorganisms at birth. However, during birth, the newborn first get in contact with microorganisms present on the vagina of mother. Usually, these are Lactobacilli. Lactobacilli colonize the intestine of the newborn. After birth, many other microbial populations begin to establish inside or on the surface of the body. These are called normal microbiota of human body. However, internal tissues of healthy human body are free of microorganisms.

A part of these microorganisms colonize on the skin, and majority enter the body and colonize on the inner surfaces such as mucous membranes of nose, throat, upper respiratory tract, intestinal tract and genitourinary tract.

A normal healthy body contains a large number of microorganisms. It has been estimated that the human body consists of 1×10^{14} microbial cells for 1×10^{13} total body cells, which means a 10 times more microbial cells than human cells.

Majority of these organisms are generally harmless or even beneficial. For example, minimal colonization of *Escherichia coli* at large intestine prevent colonization of pathogenic bacteria such as Salmonella typhi. *E. coli* in large intestine synthesizes vitamin K and some of the B vitamins that are absorbed into the bloodstream and used by body cells.

Recent interest in the importance of bacteria to human health has led to the study of probiotics. Probiotics are live microbial cultures. e.g. Yoghurt exert a beneficial effect. Serveral studies have shown that ingestion of certain lactic acid bacteria can alleviate diarrhea and prevent colonization by Salmonella enterica during antibiotic theraphy.

Although majority of human microbiota are harmless, some of them may change their interactions with human body under certain conditions and cause infections. Those microorganisms are called opportunistic pathogens. For example, E. coli is generally harmless as long as it remains in the large intestine. However, they may cause diseases, if enter other body parts (urinary bladder-urinary tract infection, lungs- pulmonary infection).

Terms related to infectious diseases

- **Pathogen:** An organisms or entity (non-living entities such as virus and prions) that is capable of causing disease
- **Host:** Organism within which infected pathogens live on or in and multiply.
- **Pathogenicity:** The ability of a pathogen to cause disease in the host by overcoming the defence of a host.
- **Parasite:** An organism or entity living on or in another living organism (host) and obtain nutrients and other resources from the host.

Characteristics of pathogenic microorganisms

- Having optimal growth conditions (e.g. temperature) that corresponds to the body conditions of the host.
- Having structures to adhere to the host cells and protect against host's defense mechanisms. e.g. capsule, pilli
- Produce toxins; endotoxins or exotoxins.
- Having enzymes for invasiveness such as phospholipase, lesithinase, and hyaluronidase.
- Having enzymes such as DNase to alter the host's metabolic processes.

Virulence and virulent factors

Microbes express their pathogenicity by means of their virulence. Virulence is the degree of pathogenicity of the pathogen. Some pathogens are highly virulent (chicken pox virus) whereas others are less virulent/ non virulent.

Few genes of pathogenic microorganisms express factors which provide them the ability to infect their host and cause disease. Such factors are called virulent factors.

The relationship between a host and a pathogen is dynamic, each modifies the activities and functions of the other. As a result, outcome of such a relationship depends on the virulence of the pathogen and the effectiveness of the host defense mechanisms.

Virulence factors enhance the pathogenicity and allows pathogen to invade and colonize host tissues and disrupt normal body function. Pathogens use two major mechanisms for pathogenicity.

1. **Invasiveness**

   It is the ability of pathogens to invade tissues by overcoming host's defense mechanisms and multiply for colonizing.

   Several extracellular enzymes produced by pathogens contribute to invasiveness.

e.g.

- Phospholipase- destroy animal cell membranes.
- Lecithinase- Hydrolyzes the lecithin component of the lipid in the cell membrane.
- Hyaluronidase- destroys the body tissue by breaking down the hyaluronic acid which is cementing substance between cells.

Pathogenic microorganisms do enter passively through various portals or natural openings such as wounds on skin, respiratory, gastrointestinal and genito-urinary tracts.

2. Toxigenecity

Ability of microorganisms to produce biochemical substances known as toxins that disrupt the normal functions of cells. These are proteins or lipopolysaccharides that produce specific harmful effects on the host, thus are called biological poisons. They may be,

- Endotoxins- Endotoxins are lipopolysaccharides. These are thermos-stable toxins which are part of the microbial cell. Toxins are released when the bacteria die and the cell wall breaks apart. All endotoxins cause the same signs of symptoms regardless of the species of pathogen. These symptoms include chills, fever, weakness, generalized aches and sometimes shock and death. Endotoxins are produced only by gram-negative bacteria

  e.g. Lipopolysaccharides of the cell walls of *Salmonella typhi*

- Exotoxins- Exotoxins are produced inside bacterial cells as part of their growth and metabolism and are secreted or released to the surrounding environment after cell lysis. Exotoxins are proteins. Majority of them are enzymes. Due to their catalytic nature even a small amount of toxin is quite harmful. These are thermo-labile protein toxins, being inactivated by boiling. Exotoxins are most commonly produced by gram-positive bacteria and a few gram negative bacteria o Exotoxins are classified into three types

- Neurotoxins- interfere with normal transmission of nerve impulses. E.g., toxins produced by *Clostridium tetani*
- Enterotoxins- stimulates cells of the gastrointestinal tract in an abnormal way. E.g. Toxins produced by *Vibrio cholera*
- Cytotoxins- kills host cells by enzymatic attack. E.g. Toxins produced by *Corynebacterium diptheriae*

Important diseases of human caused by microorganisms

Organ	Disease	Causal agent
Skin	Chickenpox Rubella Measles	Herpesvirus varicella-zoster Rubella virus Measles virus
Eye	Conjunctivitis (bacteria/virus)	Haemophilus *influenzae/Adenoviruses*
Nervous system	Bacterial meningitis	*Streptococcus pneumonia* *Haemophilus influenzae* *Neisseria meningitidis*
Nervous system	Tetanus Rabies	*Clostridium tetani* Rabies virus
Cardiovascular system	Rheumatic fever	*Streptococcus pyogenes*
Respiratory system	Tuberculosis Influenza Pneumonia	*Mycobacterium tuberculosis* Influenza virus *Streptococcus pneumoniae*
Digestive system	Hepatitis Food poisoning Cholera Typhoid	*Hepatitis A virus* Staphylococcus aureus Vibrio cholerae Salmonella typhi
Urinary system	Leptospirosis	*Leptospira interrogans*
Reproductive system	Gonorrhea Genital herpes	*Neisseria gonorrhoeae* Herpes simplex virus
Immune system	AIDS	Human immune deficiency virus (HIV)

Microbial disease control starts from avoidance and prevention of opportunities of getting infection to treatment or curative methods after infection.

Avoidance and prevention of microbial diseases

Good hygienic practices in day to day life is the best method to avoid infectious diseases. Antiseptics, disinfectants and immunization play import role in prevention of infection.

Methods of controlling microbial diseases of human

- **Use of antiseptics and disinfectants**

Antiseptics and disinfectants are chemical substances that are used to kill or reduce microbial population in order to prevent infection. However, such chemicals are not effective against some microorganisms. For example, polio virus, tuberculosis bacterium, spores of bacteria and fungi are not destroyed by most antiseptics and disinfectants.

The major difference between antiseptics and disinfectants is that antiseptics can be safely and directly applied to the human body, whereas disinfectants cannot. Therefore, antiseptics are used in disinfection of living surfaces such as skin. Disinfectants are used in disinfection of non-living surfaces such as operation theatres, bathing areas, sinks, kitchen tops, cutlery, drains etc.

Antiseptics and disinfectants are generally formulated as liquids. Their effectiveness varies with concentration, duration of exposure, temperature and presence of organic matter.

Some examples of antiseptics and disinfectants are given below.

Antiseptics: ethanol, isopropanol, chloroxylenol

Disinfectants: phenol, hypochlorites (calcium hypochlorite and sodium hypochlorite),

- **Use of Antibiotics in controlling microbial diseases**

When the body's defense fails to protect body from the infection or overcome the disease, it has to be treated by chemotherapy with antimicrobial drugs. Antimicrobial drugs kill or interfere with the growth of microorganism without damaging the host. Antibiotics are effective antimicrobial drugs against bacteria.

Some antibiotics affect against a broad range of bacteria and they are termed as broad-spectrum antibiotics, while others affect only against a specific group of bacteria and are called narrow-spectrum antibiotics.

Antibiotics show various modes of action. Some examples are given below.

- Inhibition of cell wall synthesis- Penicillin
- Inhibition of protein synthesis- Erythromycin, Tetracycline
- Disrupting plasma membrane- Daptomycin
- Inhibition of DNA/RNA synthesis- Rifampin

- **Immunization: Vaccines**

A vaccine is a suspension of weakened pathogens or fractions of organisms that is used to induce immunity. Vaccines are frequently used to control diseases caused by viruses because there is no other control methods once infected. There are several types of vaccines.

1. Live attenuated vaccines

Vaccine contains live pathogens which were deliberately weakened for its pathogenicity. These vaccines, mimic an actual infection. Since the pathogen is active inside the host, such vaccines provide lifelong immunity. More often a booster (secondary) immunization is not required. Examples for live attenuated vaccines are, vaccines for,

- Measles, Mumps and Rubella (MMR)
- Chickenpox

2. Inactivated vaccines

Pathogenic microorganism is inactivated or killed in the vaccine. Compared to live attenuated vaccines, inactivated killed vaccines often require repeated booster doses. Examples for inactivated vaccines are vaccines for,

- Virus diseases such as Rabies, Influenza, Polio
- Bacterial diseases such as Cholera

3. Subunit vaccines

Subunit vaccines contain only the antigenic fragments of a pathogen that can induce immunity in the recipient. Toxoid vaccines are the best examples for subunit vaccines, those have been used for a long time. Toxoids contain inactivated toxins derived from a pathogen. Examples for Toxoid vaccines are vaccines for Tetanus, Diphtheria, etc. Presently, subunit vaccines are produced using genetic engineering. E.g. Hepatitis-B Vaccines. Subunit vaccines usually require repeated booster dose to obtain full immunity.

Use of microorganisms in industry, agriculture and environment

Microorganisms have been exploited for various purposes long before their discovery. Babylonian and Sumerian civilizations used yeast to make alcohol in as early as 6000 BC. After the discovery of microorganisms, in the latter part of the nineteenth century, pure cultures of microorganisms are being used in the food production. This increased the understanding of microorganisms, their processes and products. At present various industries based on selected microorganisms and their qualities are in operation.

1. Advantages of using microbial processes over chemical processes

- Simple nutritional requirements are sufficient for their growth.
- They are able to convert (metabolize) a wide range of raw materials.
- They are able to convert cheap raw materials into industrially important products.
- Due to higher growth rate, they can convert the raw materials into products within a short period of time.
- Their growth conditions can be controlled to obtain desired end products.

- Reactions can be carried out at low temperatures, energy and pressures compared to the conventional industrial methods.

- They give higher yield with higher specificity when compared to the conventional industrial methods.

- Microbes are amendable to genetic manipulation to obtain desired yield and quality with high efficiency.

Basic principles of metabolic processes of microorganisms for product formations

1. Microbial cells are used as the end products. e.g. single cell proteins

2. Microbial metabolic products are used as end products- they may be either primary end products or secondary metabolites. e.g. primary end products- alcoholic beverages, secondary metabolites- antibiotics.

3. Microbial metabolic processes are used as end products. e.g. bioremediation (heavy metal remediation) , metal extraction (Cu, Fe), retting (production of fibers).

4. Genetically modified microorganisms are used to produce end products. e.g. productions of commercial enzymes (amylase from *Aspergillus niger*), vaccines (hepatitis B), hormones (insulin).

Applications of microorganisms in industries

Industrial microbiology is the large scale production of economically important products using microorganisms and their metabolic processes. Recent technological and biotechnological advances expanded the scope of industrial microbiology. Bacteria, fungi, algae and viruses are used in industry.

In industrial microbiology, microorganisms can be considered as miniature chemical factories, where various energy releasing (catabolic) and energy acquiring (anabolic) chemical reactions take place. Within this factory, raw materials (substrates) are converted into end products, one or more byproducts and wastes. End products can be separated from byproducts and wastes, by purification to obtain a purified industrial products.

Commercial products made by microorganisms and their processes

1. Single cell proteins

- Microbial cells are grown in large scale as food supplement and rich in proteins are called single cell proteins. e.g. Yeast, *Spirulina* sp. and *Chlorella* sp

$$C_6H_{12}O_6 \xrightarrow{\text{Yeast}} 2C_2H_5OH + CO_2$$

2. Alcohol and alcoholic beverages

Microorganisms are involved in the production of almost all alcoholic beverages such as beer, wine, sake, toddy and ethanol. Yeast, *Saccharomyces cerevisiae* ferment sugars into ethanol and carbon dioxide.

Globally, more than 70% of ethanol is produced by fermentation. Sucrose derived from sugarcane is the most widely used fermentation substrate. In addition, simples sugars derived from plants and dairy waste are also used

e.g. 1. Beer- by the fermentation of cereal grain malt.

2. Wine - produced by the fermentation of grapes or other suitable fruits.

3. Toddy - produced by the fermentation of sap of palms such as Palmyra and coconut.

4. Arrack- produced by fermentation of palm tree sap such as coconut and sugarcane.

3. Production of vinegar

There are two steps in vinegar production.

1. Alcoholic fermentation: sugars in malted grain, sap of palms, sugarcane and fruit juices is fermented by *S. cerevisiae*. Ethanol is subjected to acetic acid fermentation.

2. Acetic acid fermentation: Ethanol derived from alcoholic fermentation undergoes incomplete oxidation and is converted to acetic acid. This process is highly aerobic and involves *Acetobacter sp*. and *Gluconobacter sp.*

$$C_2H_5OH \xrightarrow{\text{Acetic acid bacteria}} CH_3COOH + H_2O$$

4. Dairy products

Dairy products are made by the fermentation of milk. Sugar lactose in milk is fermented by lactic acid producing bacteria into lactic acid. These bacteria are killed during pasteurization, therefore, added externally when making dairy products.

e.g.

- Curd and yoghurt is produced by fermentation of lactose sugar in milk by a mix population of *Lactobacillus bulgaricus, Lactococcus* lactis and *S. thermophilus. L. bulgaricus* add flavor and Streptococcus spp. add creamy texture and flavor.
- Production of cheese- *Streptococcus sp, Penicillium molds*
- Lactic acid is commercially produced by using waste products from cheese and butter industry. *L. bulgaricus* ferment lactose into lactic acid.

5. Organic acids

Majority of commercially produced organic acids are obtained through microbial fermentation. Fermentation substrates such as beat or cane molasses and organisms such as Aspergillus niger are used.

E.g. Citric acids- sucrose fermented by *Aspergillus niger*

6. Metal extraction

Some metals from ore are extracted with the help of microorganisms. This process is called leaching. One of the best example is extraction of copper from lower grades of copper ore for which other extraction methods are unprofitable. Thiobacillus ferrooxidans recover copper from the ore that contains iron and sulfur. About 70% of copper in the ore can be recovered by this microbial process. Uranium, gold and cobalt ore are also leached in similar manner using microbial processes.

7. Production of vitamins

Microbes can provide a source of an inexpensive source of vitamins for individual food supplements. E.g. vitamin *B12 – Pseudomonas sp*. and *Propionibacterium* sp

Riboflavin- fermentation by fungi

Vitamin C- *Acetobacter* sp

8. Vaccines

Commercial production of vaccine is done by variety of microbial antigens used in active immunization against various diseases. Some of them are genetically engineered vaccines. E.g. Hepatitis B vaccines.

Commercial production of various antibody preparation used for passive immunization. E.g. anti toxins against tetanus, Botulism toxin and immunoglobulin against rabies.

9. Enzymes

A range of enzymes are commercially produced by microorganisms. For example,

- Amylase: *Aspergillus niger, A. oryzae, Bacillus subtilis*
- Protease: *A. oryzae*
- Lipase: *Rhizopus spp.*
- Invertase: *Saccharomyces cerevisiae*
- Cellulase: *A. niger*

10. Antibiotics

Antibiotics is the most important secondary metabolites of microorganisms. Many antibiotics are still produced by microbial fermentation.

- Tetracycline: *S. aureofaciens*
- Penicillin: *Penicillium chrysogenum*
- Streptomycin: *Streptomyces griseus*

11. Hormones

a. Human insulin

Conventionally, insulin is extracted from animal pancreas. However, this is expensive and often cannot meet the demand. At present, insulin is produced cheaply by using genetically modified *E. coli* and *S. cerevisiae*. This insulin is identical to human insulin.

b. Human growth hormone

Earlier animal-derived hormones were used as alternatives to human growth hormone, but with very less efficiency. At present this hormone is successfully produced by genetically engineered E. coli in large scale.

12. Retting

Retting is the process of loosening fibers from woody stem or other plant material such as coir. For retting, plant materials are immersed in water for varying periods of time depending on the plant material. Heterogeneous bacterial populations participate in the process under aerobic or anaerobic conditions. Bacteria secrete enzymes, mainly pectinases to facilitate loosening.

13. Biogas production

Anaerobic digestion of organic waste produces various gases, called biogas. The type of biogas produced depends on the substrate biodegraded. Activity of acetogenic bacteria on the organic waste produce carbon dioxide and hydrogen, whereas, activity of methanogenic bacteria produce methane.

14. Biofuel production

Petroleum-based fuel supply is expensive and sometimes uncertain. As a result, much attention has been given to renewable replacement fuels such as ethanol, butanol, biodiesel and biogas etc. In Brazil a large amount of ethanol is produced by microbial fermentation of sugarcane to be used as a source of fuel. Also efforts have been made to produce ethanol and butanol from cellulosic materials such as wood, wastepaper and cornstalks by using genetically modified bacteria. Many researches are going on to produce biodiesel from microalgae.

15. Bakery products

Sugars in bread dough is fermented by S. cerevisiae, bakers's yeast. The primary function of fermentation in bread is to generate carbon dioxide. Bread dough is made with grain flour such as wheat, rye and rice. Dough traps carbon dioxide and rises due to the pressure during baking and form open crumb texture. .

Applications of microorganisms in environment management

Industries and agriculture release various chemicals substances that are not readily degradable in nature. For example, plastic is a synthetic substance that is not biodegraded. Residues of pesticides such as heavy metals, insecticide DDT, herbicide 2,4-D are some other examples of chemicals that are not degraded or very slowly degraded by microorganisms and retain in the soil for long periods and also contaminates groundwater.

1. Bioremediation

Bioremediation is a technology that applies of living organisms to remove, degrade or detoxify pollutants. Bioremediation naturally occurs in the soil. In most occasions microorganisms are used in bioremediation processes. Growth of microorganisms in polluted soils and water can be stimulated to promote biodegradation/ bio-removal of pollutants. Microorganisms with selected properties or genetically modified microorganisms with selected properties can be used to degrade/ remove a specific pollutant from the polluted sites. Bioremediation is currently used to,

- remediate soil and water contaminated with oil spills, toxic metal waste, hazardous organic waste etc.
- decompose wastewater from food processing and chemical plants.

2. Solid waste treatment

Accumulation of solid waste from house hold (garbage) possesses various environmental and health issues. In waste treatment, waste is degraded by microorganisms either aerobically or anaerobically. Composting, degrades waste aerobically. At the end, waste is converted to a stable material like humus.

Mostly garbage are placed as large compacted landfills or piles, where conditions are mostly anaerobic. In such situations, waste is degraded anaerobically using methanogenic bacteria. Methane gas produced as a byproduct of the degradation process. It may be used to generate electricity or as natural gas.

Applications of microorganisms in agriculture

Microorganisms have various applications in agriculture for improving yield, agronomic characteristics such as improved nitrogen and phosphorus absorption, resistance to pest and diseases and tolerance to drought etc.

1. Bio-fertilizers

Nitrogen and phosphorus are the most limiting nutrients in soil for plant growth and development. Therefore, chemical fertilizers are applied to the soil, to improve bioavailability of these nutrients. However, intensive use of synthetic fertilizers may result environmental problems such as deprivation of soil and water quality. Hence, much attention has been given for microorganisms that can be applied into the cropping systems to improve bioavailability of N and P. These microbial inoculants are called bio -fertilizers.

a. Phosphate solubilizing bacteria and mycorrhizae:

Phosphorus is the most limiting nutrient among all major plant nutrients. Bioavailability of phosphorus in any type of soil is negligible. (Very low amount of phosphorus applied to the soil remain available for plants). Phosphate solubilizing bacteria and Mycorrhizae contributes in improving the solubility of phosphorus in the soil solution. These bacteria and fungi secrete organic acids that dissolve minerals containing phosphorus and chelate cationic partners of the phosphate ions, thereby release phosphorus into the soil solution. At present, commercial formulations of microbial bio-fertilizers are available in the market.

b. Nitrogen fixing microorganism:

Biological nitrogen fixation is a process, where microorganisms convert atmospheric molecular nitrogen into its soluble form. These soluble forms of nitrogen can be assimilated by some plants directly or converted into desired soluble forms of nitrogen. e.g.

- Symbiotic nitrogen fixing
  - o *Rhizobium* sp. form intimate relationships with leguminous plants. The fixed nitrogen is released to the soil when plants die, making nitrogen available for other plants. Variousrhizobium inoculations are commercially available.
  - o Nitrogen-fixing cyanobacteria, *Anabaena* sp. form symbiotic association with water fern *Azolla*. This system has been successfully used in rice cultivation in many countries.

- Free living nitrogen fixing
    - o Free living nitrogen fixing bacteria such as Azotobacter present in high concentrations at rhizosphere.

c. Plant growth promoting bacteria

Many microorganisms in the plant rhizosphere produce plant growth-promoting substances like auxins (indole-3-acitic acid), cytokinins. These bacteria include,

Pseudomonas putida, P. fluorescens: auxin

Azotobacter sp., Rhizobium sp., B. subtilis, P. fluorescens: cytokinins

Acetobacter sp., Azospirillum sp: Gibberellin

(Names of the above microorganisms are not expected to be memorized by the students)

2. Bio-pesticides/ Bio Control Agents (BCA)

Extensive use of chemical pesticides leads to hazardous side effects to people. These substances or their residues persist in food and the environment. The residual toxicity may affect the non-target organisms. Further, over use of pesticides develop resistance against pesticides among the pests.

Therefore, environmentally friendly and less toxic alternative strategies are required to replace synthetic chemicals as pesticides. Naturally occurring microorganisms are being explored to control pests and diseases. Some microbial formulations are currently commercially available and widely used in many cropping systems. Those include, Entomopathogenic fungi, bacteria and viruses

- .Entomopathogenic fungi: These fungi infect a broad range of insects leading to insect death. They have been formulated into myco-insecticides.
- Entomopathogenic bacteria: *Bacillus thuringiensis*: insecticidal and toxic for many insect larvae. Protein crystals produced by this bacterium are toxic to larvae when ingested. This toxin is called Bt toxin. After ingestion, toxin is dissolved and lyse tissues of the guts of larvae. Majority of biopesticide formulations currently in use are Bt-based.

3. Composting

Composting is a process used to convert plant remains into the equivalent of natural humus by microbial degradation. Degradation of organic matter by a mix population of microbes under warm, moist and aerobic conditions.

Initial activity of thermophilic bacteria on the plant remains increase temperature of the pile to 55-60 °C. As a result, thermophilic bacteria dominate in the degradation process for few days. When temperature decreases over time, thermophilic microbial population is replaced by mesophilic microbial populations. The process can be enhanced by addition of moisture and supply of oxygen through turning the pile. In addition to bacteria, microbes such as fungus, Actinomycetes and protozoa also contribute to the breakdown of organic matter into compost.

Nature, distribution and role of soil microorganisms

Soil provides an adequate physical and chemical environment for growth of microorganisms in terms of space and nutrients which include minerals, decomposing organic materials, water, gases such as carbon dioxide, oxygen and nitrogen. Within a few centimeter depth of soil, there are different amounts of oxygen, moisture, light and nutrition, increasing the diversity of soil microorganisms.

The top few centimeters of the soil contains the largest community of bacteria. Microbial number declines rapidly with increasing depth. Majority of soil microflora is represented by bacteria. In addition, there are fungi, algae, protozoa and Actinomycetes. Actinomycetes, despite of being a member of domain bacteria, usually it is mentioned separately due to their significances. These microorganisms play a major role in decomposition of complex organic substances and participate in cycling of elements in biogeochemical cycles. Elements are oxidized and reduced by microorganisms for their metabolic requirements.

1. Mineralization

Mineralization is the decomposition of plant and animal residue by using extracellular enzymes of bacteria and fungi. These enzymes break down complex organic materials into simple inorganic materials such as CO_2 and H_2O. This is the major process by which plant nutrients are made available and recycling. Mineralization helps in following ways;

- helps to remove plant and animal debris from the earth surface allowing other organisms to live.
- recycle minerals which are found in limited quantities on the earth.

2. Role of microorganisms in the carbon cycle

- All organisms contain a large amount of carbon in organic compounds such as cellulose, starch, proteins and fat.
- Photosynthesis is the important first step in carbon cycle, in which, the atmospheric inorganic carbon dioxide is reduced/fixed to form organic compounds by photosynthetic organisms. Photoautotrophs such as plants, cyanobacteria, algae and photosynthetic bacteria fix carbon dioxide using energy from sunlight.
- Chemoheterotrophs such as animals and protozoa, depend on organic compounds produced by autotrophs to utilize them as their carbon source.
- Through food chain, carbon fixed from carbon dioxide by autotrophs, transferred from organism at lower trophic levels to the organisms at higher trophic levels.
- Both autotrophs and chemoheterotrophs, release a part of their fixed carbon as carbon dioxide to the atmosphere through respiration. This carbon dioxide is again made available for autotrophs.
- In chemoheterotrophs, undigested food is released to the environment as faeces which is later decomposed by soil microorganisms.

- Rest of the carbon fixed in organisms, remain within them until they die. Once the organisms are dead, these organic compounds are decomposed and carbon dioxide is returned back to the atmosphere.
- Microorganisms, mainly bacteria and fungi play a major role in organic matter decomposition.
- Microorganisms play another major role in carbon cycle in relation to methane gas- Ocean sediments contain a large amount of methane. However, about 80% of methane generated within ocean is consumed by microorganisms called methanotrophs before it reaches to the atmosphere.
- Despite of the above, methanogenic bacteria in the ocean's depths are constantly producing more methane.

3. Role of microorganisms in the nitrogen cycle

All organisms require nitrogen to synthesize protein, nucleic acid and other nitrogen-containing compounds. About 80% of molecular nitrogen available in the atmosphere. This is not biologically available for organisms. Therefore, it is essential to convert that atmospheric molecular nitrogen into bioavailable forms of nitrogen. Certain groups of microorganisms are able to fix gaseous molecular nitrogen into bioavailable forms of nitrogen such as ammonia, nitrate and nitrite. Therefore, nitrogen available on the earth, organisms and the atmosphere must follow cyclic paths.

There are four key steps in the nitrogen cycle, they are, ammonification, nitrification, denitrification and nitrogen fixation.

- **Ammonification**

More than 90% of organic nitrogen in the soil exist as proteins. Proteins from dead plants and animals are decomposed by extracellular proteolytic enzymes secreted by microorganisms into amino acids. Resulting amino acids are taken into microbial cells and are then subjected to ammonification in which, amino groups of amino acids are converted into ammonia (NH_3). In moist soil, ammonia is solubilized in water to form ammonium ions (NH_4^+). This ammonium ions are utilized by plants and soil microorganisms. Ammonia in dry soil rapidly disappear into the atmosphere.

- **Nitrification**

Nitrification is the process of oxidation of nitrogen in the ammonium ion to produce nitrate. This process is done by nitrifying bacteria living in soil in two stages.

In the first stage, microorganisms such as *Nitrosomonas*, oxidizes ammonium ions into nitrites.

$$NH_4^+ \xrightarrow{\textit{Nitrosomonas}} NO_2^-$$

In the second stage, microorganisms such as Nitrobacter oxidizes nitrites into nitrates.

$$NO_2^- \xrightarrow{\textit{Nitrobacter}} NO_3^-$$